Assignment 5 — Database Creation

Project: College Management System | Database: college_management_system | SQL dialect: MySQL 8.0

Based on the seven-table college design supplied for this project. Example records and expected outputs are illustrative, not screenshots from MySQL Workbench.

Database creation and table implementation

Run the companion College_Complete_With_Output.sql file in MySQL Workbench to create the database, insert data and display output. The DDL below implements the seven source entities.

```
CREATE DATABASE IF NOT EXISTS college_management_system;
USE college_management_system;
SET FOREIGN_KEY_CHECKS=0;
DROP TABLE IF EXISTS Result;
DROP TABLE IF EXISTS Examination;
DROP TABLE IF EXISTS Enrollment;
DROP TABLE IF EXISTS Faculty;
DROP TABLE IF EXISTS Course;
DROP TABLE IF EXISTS Student;
DROP TABLE IF EXISTS Department;
SET FOREIGN_KEY_CHECKS=1;
CREATE TABLE Department (Department_ID INT PRIMARY KEY, Department_Name VARCHAR(100) NOT
NULL UNIQUE, Location VARCHAR(100) NOT NULL, HOD VARCHAR(100));
CREATE TABLE Student (Student_ID INT PRIMARY KEY, Name VARCHAR(100) NOT NULL, Email
VARCHAR(100) NOT NULL UNIQUE, Phone VARCHAR(15), Department_ID INT NOT NULL REFERENCES
Department(Department_ID), Admission_Year INT NOT NULL CHECK(Admission_Year BETWEEN 2000
AND 2100));
CREATE TABLE Course (Course_ID INT PRIMARY KEY, Course_Name VARCHAR(100) NOT NULL, Credits
INT NOT NULL CHECK(Credits BETWEEN 1 AND 6), Department_ID INT NOT NULL REFERENCES
Department(Department_ID), UNIQUE(Course_Name,Department_ID));
CREATE TABLE Faculty (Faculty_ID INT PRIMARY KEY, Name VARCHAR(100) NOT NULL, Email
VARCHAR(100) NOT NULL UNIQUE, Phone VARCHAR(15), Department_ID INT NOT NULL REFERENCES
Department(Department_ID));
CREATE TABLE Enrollment (Enrollment_ID INT PRIMARY KEY, Student_ID INT NOT NULL REFERENCES
Student(Student_ID), Course_ID INT NOT NULL REFERENCES Course(Course_ID), Semester INT NOT
NULL CHECK(Semester BETWEEN 1 AND 8), Enrollment_Date DATE NOT NULL DEFAULT (CURRENT_DATE),
UNIQUE(Student_ID,Course_ID,Semester));
CREATE TABLE Examination (Exam_ID INT PRIMARY KEY, Course_ID INT NOT NULL REFERENCES
Course(Course_ID), Exam_Date DATE NOT NULL, Exam_Type VARCHAR(50) NOT NULL CHECK(Exam_Type
IN ('Midterm','Final','Quiz')), Total_Marks INT NOT NULL CHECK(Total_Marks > 0));
CREATE TABLE Result (Result_ID INT PRIMARY KEY, Exam_ID INT NOT NULL REFERENCES
Examination(Exam_ID), Student_ID INT NOT NULL REFERENCES Student(Student_ID), Marks INT NOT
NULL CHECK(Marks >= 0), Grade VARCHAR(5) NOT NULL CHECK(Grade IN ('A+','A','B','C','D',
'F')), UNIQUE(Exam_ID,Student_ID));
```

Relationship overview

Department 1:N Student, Faculty and Course; Student 1:N Enrollment; Course 1:N Enrollment and Examination; Examination 1:N Result; Student 1:N Result. Enrollment resolves the many-to-many relationship between Student and Course.

Verification

```
SHOW TABLES;
DESCRIBE Student;
DESCRIBE Enrollment;
SHOW CREATE TABLE Result;
```